

Computer Vision Quiz on HOG

Google Form link:

<https://docs.google.com/forms/d/e/1FAIpQLSeHXEjjPGGz8SvvjnP4bCWEslbMk9XfPLY6k0YrLfN0uNH4nw/viewform>

Question 1

Why b is correct: HOG simplifies an image by extracting essential structural information, focusing on where pixel intensity changes sharply to identify edges and outlines.

Why a, c, d are incorrect: (Ans a and c) It is common to think image recognition requires full color detail but HOG intentionally discards color and shadows because they fluctuate wildly with lighting. (Ans d) HOG acts as a descriptor for machine learning models, not a replacement for them.

Question 2

Why c is correct: Lighting, shadows, and noise change raw pixel numbers completely even if the object's physical shape remains the same.

Why a, b, d are incorrect: (Ans a and d) One might assume digital grids are perfectly stable or that computers struggle to read them. (Ans b) Computers easily read pixel grids, but the issue is their wild fluctuation under different environments, making them unreliable for shape detection.

Question 3

Why c is correct: Gradients measure the rate of intensity change which effectively highlights boundaries that define form.

Why a, b, d are incorrect: The distractors represent actual, important steps in the HOG pipeline, making them easy to confuse. However, grouping pixels (Ans a), normalizing brightness (Ans d), and flattening vectors (Ans b) happen after gradients are computed.

Question 4

Why b is correct: An 8x8 pixel grid is the standard local region used to summarize edge patterns.

Why a, c, d are incorrect: (Ans a and c) These are all actual dimensions used elsewhere in HOG. (Ans a and c) 2x2 and 16x16 refer to the sizes of blocks of cells, while (Ans d) 64x128 is the size of the entire detection window.

Question 5

Why a is correct: The standard HOG splits the 180 degree range into 9 distinct bins.

Why b, c, d are incorrect: (Ans c) 1 bin is too small and it would capture little to no information, while (Ans b and d) 180 and 20 bins are a lot and they would capture a lot of noise

Question 6

Why a is correct: Block Normalization rescales gradients so the descriptor reflects relative edge patterns rather than absolute brightness.

Why b, c, d are incorrect: It's easy to confuse normalization with basic image editing. (Ans d) While grayscale conversion does happen, it's done during early preprocessing. Normalization adjusts mathematical vector values, it doesn't change pixel counts (Ans b) or removes edges (Ans c).

Question 7

Why d is correct: The Sobel filters slide across the image to estimate intensity changes

Why a, b, c are incorrect: These are all real mathematical terms used in the HOG pipeline. (Ans a) Gamma is for color pre-processing, (Ans b) SVM is the final classifier, not the gradient operator and (Ans c) L2-norm is for block normalization

Question 8

Why c is correct: Stronger edges (higher magnitude) get a proportionally Louder voice in the histogram.

Why a, b, d are incorrect: Spatial positioning is a key concept in HOG, so it's easy to mistake it for (Ans b) position or (Ans d) distance . However, position only dictates which cell the pixel votes in, while magnitude dictates the weight of that vote. (Ans a) Color is entirely discarded.

Question 9

Why b is correct: Grouping pixels into 8x8 cells provides Spatial Invariance, making the descriptor robust to small geometric shifts.

Why a, c, d are incorrect: One might incorrectly assume HOG uses (Ans a) motion or (Ans c and d) color tracking to solve this. Instead, it relies purely on the mathematical summation within local cells to smooth out slight misalignments.

Question 10

Why b is correct: It is because blocks overlap, a single cell is normalized multiple times relative to different neighbors which provides a redundant and deeply context rich representation.

Why a, c, d are incorrect: (Ans and d) 1-D kernels and SVM classifiers are real components of the system, but they are standard machine learning tools, not the unique architectural trick that gives HOG its density. (Ans c) Star-shaped icons are merely visual aids for humans, not part of the algorithm itself.